

Education

- 2018 – 2020 **M.Sc. in Mathematics, IMPA.** *Rio de Janeiro, Brazil.*
- 2014 – 2018 **B.Sc. in Mathematics, Universidad de los Andes.** *Bogotá, Colombia*
- Minor in Literature.
 - Minor in History of Music.
- 2025 **BlueDot Impact** (San Francisco). Two certifications (q.v. bluedot.org):
- *AGI Strategy.*
 - *Technical AI Safety.*

Experience

Selected experience, ordered by professional relevance.

- 2024, 2025 **Researcher at CHAI, UC Berkeley** (twice). *Berkeley's Center for Human-Compatible AI, led by Stuart Russell.*
- 2025: used reinforcement learning to train an assistant for a cooperative Minecraft-like game whose goal is hidden and must be inferred from human actions (with Cameron Allen).
 - 2024: the AI-alignment "Asking for Help" program (with Benjamin Plaut), which led to the paper co-authored with Stuart Russell.
- 2025, 2026 **Lecturer, Universidad de los Andes.** *School of Economics.* Instructor for two advanced courses:
- "Machine Learning for Business Intelligence".
 - "Deep Learning and Advanced Machine Learning Applications".
- 2023 – 2025 **ML Researcher and Python Developer, Quantil.** A leading Colombian applied-mathematics and AI firm, with clients such as Ecopetrol, Banco W and the Colombian government.
- 2018, 2021, 2022 **Curriculum Editor and Translator, Illustrative Mathematics.** A leading math-education organization whose evidence-based curricula are used internationally.
- 2016, 2017 **Teaching Assistant, Universidad de los Andes.** *Department of Mathematics.* Two introductory courses:
- Linear Algebra.
 - Integral Calculus.
- 2023 **Research Assistant, UCLA.** Qualitative data analysis and conceptual editing for a social-welfare project with teenage mothers in Cartagena and Medellín.
- 2022 – present **Data Consultant, Joyería Liévano.** Predictive models and customer analytics.

Honors, Projects and Invited Talks

- 2025 **First author at NeurIPS.** *One of the world's leading conferences in artificial intelligence.* Presented the paper "*Echo of Bayes: Learned Memory Functions Can Recover Belief States*" (with UC Berkeley and Brown University). Available on OpenReview.

- 2025 **Co-author of a paper with Stuart Russell.** (Renowned scientist and author of *Artificial Intelligence: A Modern Approach*, the world's most widely used AI textbook.) Paper: "*Safe Learning Under Irreversible Dynamics via Asking for Help*" (arXiv:2502.14043).
- 2026 **Paper in progress** on recommender systems, with Sana Panday (UC Berkeley): designing recommendation algorithms that build in peace-building incentives rather than optimizing solely for attention capture (*engagement*).
- 2018 **Full scholarship with stipend (CNPq)** for the M.Sc. at IMPA.
- 2024, 2025, 2026 **Invited participant at the annual CHAI workshop** in Asilomar, California. An invitation-only retreat that each year brings together about 200 leading researchers in AI and AI safety.

Skills

- Python (Pandas for data science; JAX/PyTorch for reinforcement learning).
- Strong background in statistics, probability, linear algebra and formal logic.
- Mathematics pedagogy: curriculum design (Illustrative Mathematics) and university teaching (Uniandes).
- Infrastructure and data: Azure (OCR, Cognitive Search, Cosmos DB), Databricks, $\text{\LaTeX}$.

Languages

Spanish, *Native*.

English, *Highly proficient*.

Portuguese, *Advanced*.